SQL Practice Exercise: Joins, UNION, Filtering, and Aggregates

EXERCISE 4 SQL Joins

1. Retrieve all employees and their assigned projects using an **INNER JOIN**.

Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status.

```
SELECT *

FROM employees AS A

INNER JOIN projects AS B

ON A.Employees_id = B.Employees_id;
```

EmlD	FiName	LName	Department	Salary	PrjID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Dept	10000	Completed
2	Alice	Smith	HR	60000	102	Emp Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Complete
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

2. Retrieve all employees and their assigned projects, including employees who have no projects using a **LEFT JOIN**.

Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status.

```
SELECT *
FROM employees AS A
LEFT JOIN projects AS B
ON A.Employees_id = B.Employees_id;
```

EmpID	FirstName	LastName	Dept	Salary	ProjectID	P_name	Budget	Status
1	John	Doe	IT	70000		AI Development	100000	C
2	Alice	Smith	HR	60000		Employee Training	50 000	O
3	Bob	Johnson	FinC	75000		Cybersecurity Audit	75 000	P
4	David	Brown	IT	72000		Financial Analysis	90 000	O

5	Emma	Wilson	Sales	65000	Market Expansion	65 000	C
6	Michael	Clark	Finance	80000	Risk Management	80 000	P

3. Retrieve all projects and their assigned employees, including projects that have no employees using a **RIGHT JOIN**.

Return: ProjectID, ProjectName, Budget, Status, EmployeeID, FirstName, LastName, Department, Salary.

```
SELECT *
FROM employees AS A
RIGHT JOIN projects AS B
ON A.Employees_id = B.Employees_id;
```

4. Retrieve all employees and projects, including those without a match in either table using a **FULL OUTER JOIN.**

o Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status.

```
SELECT *
FROM employees AS A
FULL OUTER JOIN projects AS B
ON A.Employees_id = B.Employees_id;
```

Emp ID	FiName	LName	Department	Salary	PrjID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Com
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	On
4	David	Brown	IT	72000	NULL	NULL	NULL	N
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	C
6	Michael	Clark	Finance	80000	106	Risk Man	80000	P